# Assignment 1 (10% of total marks)

**Due date:** Sunday, 26 April 2026 by 9:00 pm Singapore time.

**Scope:**

The tasks of this assignment cover **functional dependency**. The assignment covers the topics discussed in lecture 1, and 2.

**Assessment criteria:**

Marks will be awarded for:
- Correct,
- Comprehensive, and
- Appropriate

application of the materials covered in this subject.

A late submission penalty (5% of the total mark) will be applied for every 24 hours late.

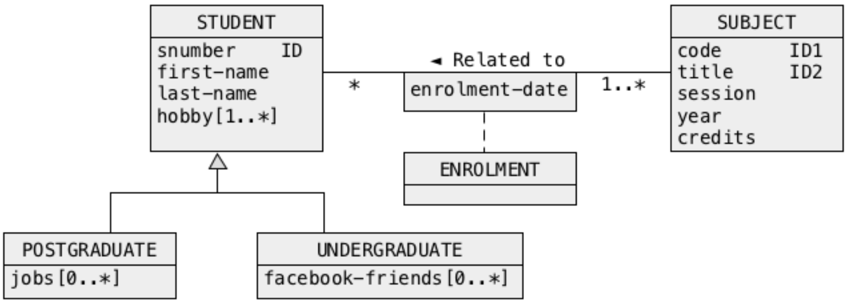
Submission must be done online by using the submission link associated with assignment 1 for this subject on MOODLE. The size limit for all submitted materials is 20MB.

**Note: Only one submission is allowed.** Ensure you upload the correct files with the exact required contents before the deadline. If you need an extension (maximum 1 week), you must submit an Academic Consideration request via SOLS **before or on the due date**. Requests submitted after the due date are likely to be rejected.

**Assignment Specification:**

**Task 1 (3.0 marks)**
**Analysis of relational schemas and normalization**

Consider the following conceptual schema of a sample database domain that contains information about undergraduate and postgraduate students, subjects, and enrolments performed by the students.




A database designer applied a process of logical design and he/she transformed the conceptual schema into the following collection of relational schemas.

```
STUDENT(snumber,first-name,last-name,hobby,jobs,facebook-friends)
primary key = (snumber)

SUBJECT(code,title,session,year,credits)
primary key = (code)
candidate key = (title)

ENROLMENT(snumber,code,title,enrolment-date)
primary key = (snumber,code, enrolment-date)
foreign key 1 = (snumber) references STUDENT(snumber)
foreign key 2 = (code) references SUBJECT(code)
foreign key 3 = (title) references SUBJECT(title)
```

(1) Find all functional and multivalued dependencies in the relational schemas STUDENT, SUBJECT, and ENROLMENT.

STUDENT
snumber → first-name,last-name
snumber →→ hobby | jobs | facebook-friends

SUBJECT
code → title,session,year,credits
title → code,title,session,year,credits
title → code
code → title

ENROLMENT
snumber,code,enrolment-date → title
snumber,title,enrolment-date → code
title → code
code → title

(2) Find all minimal keys in the relational schemas STUDENT, SUBJECT, and ENROLMENT. List the derivations of all minimal keys.

STUDENT
If snumber → first-name,last-name then
snumber,hobby,jobs, facebook-friends → first-name,last-name
minimal key = (snumber,hobby,jobs,facebook-friends)

SUBJECT
code → title,session,year,credits
minimal key 1 = (code)
title → code,title,session,year,credits
minimal key 2 = (title)

ENROLMENT
snumber,code,enrolment-date → title
minimal key 1 = (snumber,code,enrolment-date)
snumber,title,enrolment-date → code
minimal key 1 = (snumber,title,enrolment-date)

(3) For each one of the relational schemas find the highest normal form a schema is in. List the justifications for each highest normal form found.

STUDENT
snumber → first-name,last-name
minimal key = (snumber,hobby,jobs,facebook-friends)
snumber → first-name,last-name
not 2NF: nonprime attributes first-name,last-name depend on a subset (snumber) of o minimal key.

SUBJECT
code → title,session,year,credits
minimal key 1 = (code)
title → code,title,session,year,credits

minimal key 2 = (title)
title → code
code → title
BCNF: left hand side of each FD is a superkey.

ENROLMENT
snumber,code,enrolment-date → title
minimal key 1 = (snumber,code,enrolment-date)
snumber,title,enrolment-date → code
minimal key 1 = (snumber,title,enrolment-date)
title → code
code → title
3NF: for each FD its left hand side is a superkey or right hand side is prime.
not BCNF: left hand side of title → code, code → title is not a superkey.

(4) Decompose all relational schemas that are not in 4NF into 4NF. List all relational schemas obtained from the decompositions.

Decomposition of STUDENT to 4NF:
STUDENT'(snumber,first-name,last-name)
SHOBBY(snumber,hobby)
SJOBS(snumber,jobs)
SFRIENDS(snumber,facebook-friends)

Decomposition of SUBJECT to 4NF is not needed because the schema is in BCNF and there are no nontrivial multivalued dependencies.

Decomposition of ENROLMENT to 4NF:

ENROLMENT'(snumber,code,enrolment-date), BCNF because there are no nontrivial FDs, 4NF because there are no nontrivial multivalued dependencies.
CT(code,title) 4NF because a schema consists of two attributes.

Deliverables
A file solution1.pdf with the outcomes of the steps (1), (2), (3), and (4) listed above. Note, that "educated guesses" of the solutions score no marks. You must provide the complete justifications of your answers.

Submission of a file with a different name and/or different extension and/or different type scores no marks

## Task 2 (3.0 marks)
## Analysis of relational schemas and normalization

Consider the relational schemas given below and the respective sets of functional dependencies valid in the schemas.

$R(P, Q, R, S, T, U, V, W)$
$Functional\ Dependency: RW \rightarrow V, P \rightarrow QR, Q \rightarrow RUW, T \rightarrow P, U \rightarrow TV$

(i) Find **all** the minimal super keys of the relational table $R$. List the derivations of all minimal keys.
(ii) Identify the highest normal form of the relational table $R$. Remember that the identification of a normal form requires analysis of the valid functional dependencies.
(iii) Decompose the relational table $R$ into minimal number of normalized relational tables in BCNF. Remember to indicate the primary key and foreign keys (if any).

**Deliverables**
A file **solution1.pdf** with the outcomes of the steps (i), (ii), and (iii) listed above. Note, that "educated guesses" of the solutions score no marks. You must provide the complete justifications of your answers.

Submission of a file with a different name and/or different extension and/or different type scores no marks!

Sample solution includes but is not limited to the following:

(i) Find all the minimal super key of the relational table $R$. List the derivations of all minimal keys.

$R(P, Q, R, S, T, U, V, W)$

$Functional\ Dependency:$
$R, W \rightarrow V,$
$P \rightarrow Q, R$
$Q \rightarrow R, U, W$
$T \rightarrow P$
$U \rightarrow T, V$

Closure of attribute $\{R, W\}^+$:

- $\{R, W\}^+ = \{R, W\}$
  $= \{R, W, V\}$

Closure of attribute $\{P\}^+$:

- $\{P\}^+ = \{P\}$
  $= \{P, Q, R\}$
  $= \{P, Q, R, U, W\}$
  $= \{P, Q, R, U, W, T, V\}$

Closure of attribute $\{Q\}^+$:

- $\{Q\}^+ = \{Q\}$
  $= \{Q, R, U, W\}$
  $= \{Q, R, U, W, T, V\}$
  $= \{Q, R, U, W, T, V, P\}$

Closure of attribute $\{T\}^+$:

- $\{T\}^+ = \{T\}$
  $= \{T, P\}$
  $= \{T, P, Q, R\}$
  $= \{T, P, Q, R, U, W\}$
  $= \{T, P, Q, R, U, W, V\}$

Closure of attribute $\{U\}^+$:

- $\{U\}^+ = \{U\}$
  $= \{U, T, V\}$
  $= \{U, T, V, P\}$
  $= \{U, T, V, P, Q, R\}$
  $= \{U, T, V, P, Q, R, W\}$

From the closure of attributes found above, we have the following functional dependencies established:

- $(R, W) \rightarrow V$
- $P \rightarrow Q, R, U, W, T, V$
- $Q \rightarrow R, U, W, T, V, P$
- $T \rightarrow P, Q, R, U, W, V$
- $U \rightarrow T, V, P, Q, R, W$

(1):

- $(R, W) \rightarrow V$ is a valid functional dependency.
- $P \rightarrow P\ is\ a\ trivial\ dependency.$
- $Q \rightarrow Q\ is\ a\ trivial\ dependency.$
- $S \rightarrow S\ is\ a\ trivial\ dependency.$
- $T \rightarrow T\ is\ a\ trivial\ dependency.$
- $U \rightarrow U\ is\ a\ trivial\ dependency.$
- $V \rightarrow V\ is\ a\ trivial\ dependency.$

Through Armstrong's augmentation axiom, we have

- $(R, W, P, Q, S, T, U) \rightarrow V$
- The functional dependency $(R, W, P, Q, S, T, U) \rightarrow V$ covers the entire relational schema $R$, then its left-hand side $(R, W, P, Q, S, T, U)$ is a super key.

(2):
- $P \rightarrow Q, R, U, W, T, V$ is a valid functional dependency.
- $S \rightarrow S \, is \, a \, trivial \, dependency.$

Through Armstrong's augmentation axiom, we have
$(P, S) \rightarrow Q, R, U, W, T, V$

The functional dependency $(P, S) \rightarrow Q, R, U, W, T, V$ covers the entire relational schema $R$, then its left-hand side $(P, S)$ is a super key.

(3):
- $Q \rightarrow R, U, W, T, V, P$ is a valid functional dependency.
- $S \rightarrow S \, is \, a \, trivial \, dependency.$

Through Armstrong's augmentation axiom, we have
$(Q, S) \rightarrow R, U, W, T, V, P$

The functional dependenc $(Q, S) \rightarrow R, U, W, T, V, P$ covers the entire relational schema $R$, then its left-hand side $(Q, S)$ is a super key.

(4):
- $T \rightarrow P, Q, R, U, W, V$
- $S \rightarrow S \, is \, a \, trivial \, dependency.$

Through Armstrong's augmentation axiom, we have
$(T, S) \rightarrow P, Q, R, U, W, V$

The functional dependency $(T, S) \rightarrow P, Q, R, U, W, V$ covers the entire relational schema $R$, then its left-hand side $(T, S)$ is a super key.

(5):
- $U \rightarrow T, V, P, Q, R, W$
- $S \rightarrow S \, is \, a \, trivial \, dependency.$

Through Armstrong's augmentation axiom, we have
$(U, S) \rightarrow T, V, P, Q, R, W$

The functional dependency $(U,S) \rightarrow T,V,P,Q,R,W$ covers the entire relational schema $R$, then its left-hand side $(U,S)$ is a super key.

There are five super keys identified, they are $(R,W,P,Q,S,T,U)$, $(P,S)$, $(Q,S)$, $(T,S)$, and $(U,S)$. Since $(P,S)$, $(Q,S)$, $(T,S)$, and $(U,S)$ are subsets of $(R,W,P,Q,S,T,U)$, hence, $(P,S)$, $(Q,S)$, $(T,S)$, and $(U,S)$ are the minimal super keys of the relational table $R$.

(ii) Identify the highest normal form of the relational table $R$. Remember that the identification of a normal form requires analysis of the valid functional dependencies.

- $(P,S) \rightarrow Q,R,U,W,T,V$

  From the above functional dependency, there exists a partial dependency $P \rightarrow Q,R,U,W,T,V$, and an existence of a functional dependency is a violation of the requirement to be in second normal form (2NF). Hence the relational table $R$ is in **first normal form (1NF).**

- $(Q,S) \rightarrow R,U,W,T,V,P$

  From the above functional dependency, there exists a partial dependency $Q \rightarrow R,U,W,T,V,P$, and an existence of a functional dependency is a violation of the requirement to be in second normal form (2NF). Hence the relational table $R$ is in **first normal form (1NF).**

- $(T,S) \rightarrow P,Q,R,U,W,V$

  From the above functional dependency, there exists a partial dependency $(T) \rightarrow P,Q,R,U,W,V$, and an existence of a functional dependency is a violation of the requirement to be in second normal form (2NF). Hence the relational table $R$ is in **first normal form (1NF).**

- $(U,S) \rightarrow T,V,P,Q,R,W$

  From the above functional dependency, there exists a partial dependency $U \rightarrow T,V,P,Q,R,W$, and an existence of a functional dependency is a violation of the requirement to be in second normal form (2NF). Hence the relational table $R$ is in **first normal form (1NF)**.

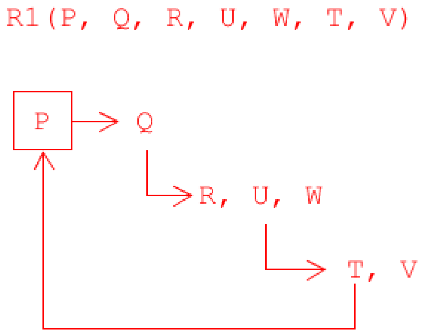
(iii) Decompose the relational table $R$ into minimal number of normalized relational tables in BCNF. Remember to indicate the primary key and foreign keys (if any).

- $(P,S) \rightarrow Q,R,U,W,T,V$

There exists a partial functional dependency $P \rightarrow Q, R, U, W, T, V$ from the functional dependency $(P, S) \rightarrow Q, R, U, W, T, V$, and we have the following:

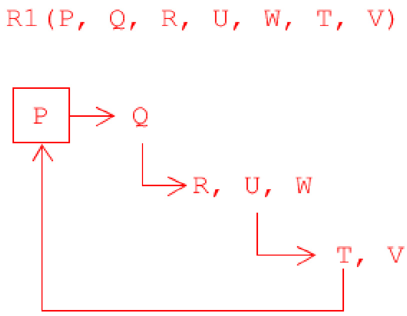
$R1(P, Q, R, U, W, T, V)$
$PK: P$

$R2(P, S)$
$PK: (P, S)$
$FK: P\ references\ R1(P)$




The relational schema R1 contains transitive dependencies Q->R, U, W, and U->T, V. In addition, R1 also contains a non-trivial dependency T->P. Hence, the relational schema R1 is in second normal form (2NF).

To bring the relational R1 to third normal form (3NF), we remove all the transitive dependencies. After removing all the transitive dependencies, we have the following:




```
R13(U, T, V)
PK: U

R12(Q, R, U, W)
PK: Q
FK: U references R13(U)

R11(P, Q)
PK: P
FK: Q references R12(Q)
```

$R2(P, S)$
$PK: (P, S)$
$FK: P\ references\ R11(P)$

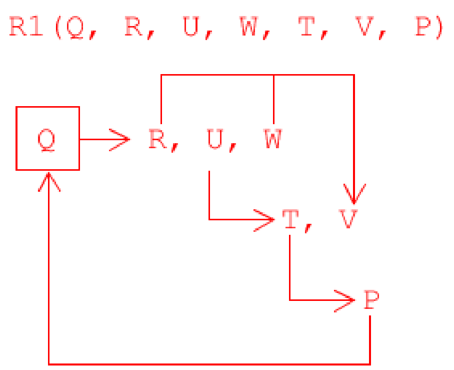
The relational schemas R2, R11, R12, and R13 do not have transitive dependency as well as non-trivial dependency. Hence, the relational schemas are now in Boyce-Codd normal form (BCNF).

- $(Q, S) \rightarrow R, U, W, T, V, P$

  There exists a partial functional dependency $Q \rightarrow R, U, W, T, V, P$ from the functional dependency $(Q, S) \rightarrow R, U, W, T, V, P$, and we have the following:

  $R1(Q, R, U, W, T, V, P)$
  $PK: Q$

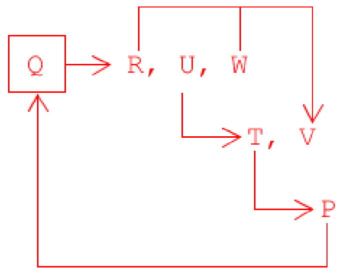
  $R2(Q, S)$
  $PK: (Q, S)$
  $FK: Q\ references\ R1(Q)$




The relational schema R1 contains transitive dependencies RW-> V, U-> T, V, and T-> P. In addition, R1 also contains a non-trivial dependency P-> Q. Hence, the relational schema R1 is in second normal form (2NF).

To bring the relational R1 to third normal form (3NF), we remove all the transitive dependencies. After removing all the transitive dependencies, we have the following:

```
R1(Q, R, U, W, T, V, P)
```




```
R14(T, P)
PK: T

R13(U, T, V)
PK: U
FK: T references T14(T)

R12(R, W, V)
PK: RW

R11(Q, R, U, W)
PK: Q
FK1: RW references R12(RW)
FK2: U references R13(U)

R2(Q, S)
PK: (Q, S)
FK: Q references R11(Q)
```

The relational schemas R2, R11, R12, R13, and R14 do not have transitive dependency as well as non-trivial dependency. Hence, the relational schemas are now in Boyce-Codd normal form (BCNF).

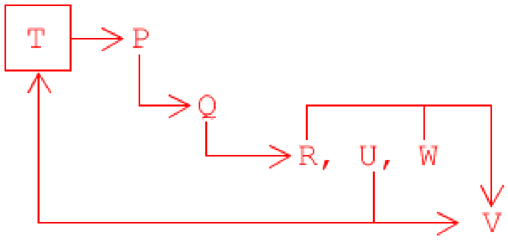
- $(T, S) \rightarrow P, Q, R, U, W, V$

  There exists a partial functional dependency $T \rightarrow P, Q, R, U, W, V$ from the functional dependency $(T, S) \rightarrow P, Q, R, U, W, V$, and we have the following:

  $R1(T, P, Q, R, U, W, V)$
  $PK: T$

  $R2(T, S)$
  $PK: (T, S)$
  $FK: T\ references\ R1(T)$

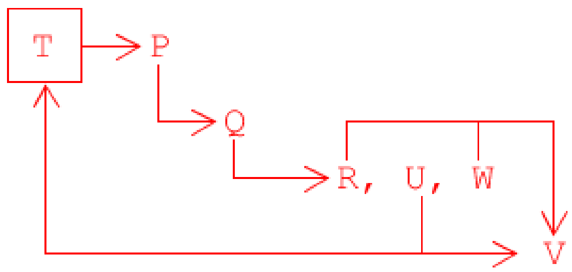
R1(T, P, Q, R, U, W, V)




The relational schema R1 contains transitive dependencies P-> Q, Q-> R, U, W, (R, W)-> VV, U-> T, V, and T-> P. In addition, R1 also contains a non-trivial dependency P-> Q. Hence, the relational schema R1 is in second normal form (2NF).

To bring the relational R1 to third normal form (3NF), we remove all the transitive dependencies. After removing all the transitive dependencies, we have the following:

R1(T, P, Q, R, U, W, V)




R1(T, P)
PK: T

R14(U, T, V)
PK: U
FK: T references R1(T)

R13(R, W, V)
PK: (R, W)

R12(Q, R, U, W)
PK: Q
FK1: (R, W) references R13(R, W)
FK2: U references R14(U)

R11(P, Q)
PK: P
FK: Q references R12(Q)

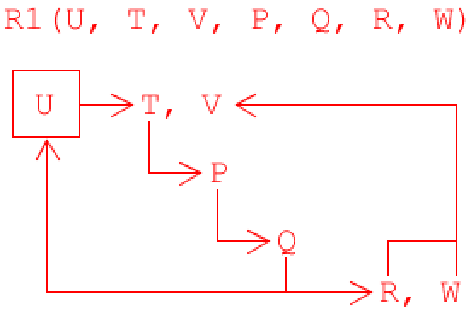
R2(T, S)
PK: (T, S)
FK: T references R1(T)

- $(U, S) \rightarrow T, V, P, Q, R, W$

  There exists a partial functional dependency $U \rightarrow T, V, P, Q, R, W$ from the functional dependency $(U, S) \rightarrow T, V, P, Q, R, W$, and we have the following:

  $R1(U, T, V, P, Q, R, W)$
  $PK: U$

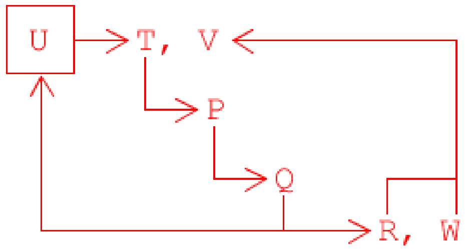
  $R2(U, S)$
  $PK: (U, S)$
  $FK: U\ references\ R1(U)$




The relational schema R1 contains transitive dependencies T-> P, P-> Q, Q-> R, U, W, and (R, W)-> V. In addition, R1 also contains a non-trivial dependency Q-> U. Hence, the relational schema R1 is in second normal form (2NF).

To bring the relational R1 to third normal form (3NF), we remove all the transitive dependencies. After removing all the transitive dependencies, we have the following:

R1(U, T, V, P, Q, R, W)




R11(U, T)
PK: U

R15(R, W, V)
PK: (R, W)

R14(Q, R, U, W)
PK: Q
FK1: (R, W) references R15(R, W)
FK2: U references R11(U)

R13(P, Q)
PK: P
FK: Q references R14(Q)

R12(T, P)
PK: T
FK: P references R13(P)

R2(U, S)
PK: (U, S)
FK: U references R11(T)

**Task 2 (3.0 marks)**
**Analysis of relational schemas and normalization**

A book is described by a catalog number, a book title, author's first name, author's last name, a publisher name, the year the book was published, and a price of the book. An author may write many books and each book may be written by one or more authors. The information is stored in a relational table book as follow:

BOOK (CatalogNum, BookTitle, AuthorFName, AuthorLName, PublisherName, YearOfPublication, Price)

The following dependencies exist in the relational table BOOK:

- $CatalogNum \rightarrow BookTitle, PublisherName, YearOfPublication$
- $BookTitle, PublisherName, YearOfPublication \rightarrow Price$
- $AuthorFirstName, AuthorLastName, BookTitle \rightarrow CatalogNum$

(i) Find all the minimal super key of the relational table BOOK. List the derivations of all minimal keys.
(ii) Identify the highest normal form of the relational table BOOK. List the justifications for each highest normal form found.
(iii) Decompose the relational table BOOK into minimal number of relational tables in BCNF. List all relational tables obtained from the decompositions.

Sample solution includes but is not limited to the following:

(i) Find all the minimal super key of the relational table BOOK. List the derivations of all minimal keys.

- If $CatalogNum \rightarrow BookTitle, PublisherName, YearOfPublication$ and $BookTitle, PublisherName, YearOfPublication \rightarrow Price$, then through transitivity axiom $CatalogNum \rightarrow Price$.
- Since an author may write many books and a book may be written by many authors, through augmentation axiom, $CatalogNum, AuthorFirstName, AuthorLastName \rightarrow BookTitle, PublisherName, YearOfPublication, Price, AuthorFirstName, AuthorLastName$.
- If $CatalogNum, AuthorFirstName, AuthorLastName \rightarrow BookTitle, PublisherName, YearOfPublication, Price, AuthorFirstName, AuthorLastName$ is valid in the relational table BOOK, and it covers the entire relational schema, then the left-hand-side of the functional dependency ($\boldsymbol{CatalogNum, AuthorFirstName, AuthorLastName}$) is a minimal super key.

- If $AuthorFirstName, AuthorLastName, BookTitle \rightarrow CatalogNum$, and $CatalogNum \rightarrow BookTitle, PublisherName, YearOfPublication$, then through transitivity axiom, $AuthorFirstName, AuthorLastName, BookTitle \rightarrow CatalogNum, BookTitle, PublisherName, YearOfPublication$.
- Similarly, if $BookTitle, PublisherName, YearOfPublication \rightarrow Price$, then through transitivity axiom, $AuthorFirstName, AuthorLastName, BookTitle \rightarrow CatalogNum, BookTitle, PublisheName, YearOfPublication, Price$.
- If $AuthorFirstName, AuthorLastName, BookTitle \rightarrow CatalogNum, BookTitle, PublisherName, YearOfPublication, Price$ is valid in the relational table BOOK, and it covers the entire relational schema, then the left-hand-side of the functional dependency $(\boldsymbol{AuthorFirstName}, \boldsymbol{AuthorLastName}, \boldsymbol{BookTitle})$ is a minimal super key.

(ii) Identify the highest normal form of the relational table BOOK. List the justifications for each highest normal form found.

- Since in the functional dependency $CatalogNum, AuthorFirstName, AuthorLastName \rightarrow BookTitle, PublisherName, YearOfPublication, Price, AuthorFirstName, AuthorLastName$, there exist a partial dependency $CatalogNum \rightarrow BookTitle, PublisherName, YearOfPublication$. The relational table BOOK has a partial dependency violation. Hence, the highest normal form of the relational table BOOK is in First Normal Form (1NF).

- If $AuthorFirstName, AuthorLastName, BookTitle \rightarrow CatalogNum, BookTitle, PublisherName, YearOfPublication, Price$ is considered, then there a transitive dependency $AuthorFirstName, AuthorLastName, BookTitle \rightarrow CatalogNum$ and $CatalogNum \rightarrow PublisherName, YearOfPublication$. This makes the relational table BOOK not qualified to be in 3NF. In addition, there exist a non-trivial functional dependency $CatalogNum \rightarrow BookTitle$. This further make the relational table BOOK to violate the requirement to be in BCNF. Hence, the highest normal form of the relational table BOOK is in Second Normal Form (2NF).

(iii) Decompose the relational table BOOK into minimal number of relational tables in BCNF. List all relational tables obtained from the decompositions.

In the first scenario, $CatalogNum, AuthorFirstName, AuthorLastName \rightarrow BookTitle, PublisherName, YearOfPublication, Price, AuthorFirstName, AuthorLastName$, to normalize the relational table BOOK into BCNF, we first remove the partial dependency from the relational table BOOK, and decompose the relational table into:

$BOOKCATALOG2NF(CatalogNum, BookTitle, PublisherName, YearOfPublication, Price)$
$PK: CatalogNum$

$AUTHORCATALOG2NF(CatalogNum, AuthorFirstName, AuthorLastName)$
$PK: (CatalogNum, AuthorFirstName, AuthorLastName)$
$FK: (CatalogNum)\ references\ BOOKCATALOG2NF(CatalogNum)$

Next, bring the relational table BOOKCATALOG2NF to 3NF by removing the transitive dependency $BookTitle, PublisherName, YearOfPublicatiion \rightarrow Price$, and we have the following:

$BOOKPRICE3NF(BookTitle, PublisherName, YearOfPublication, Price)$
$PK: (BookTitle, PublisherName, YearOfPublication)$

$BOOKCATALOG3NF(CatalogNum, BookTitle, PublisherName, YearOfPublication)$
$PK: CatalogNum$
$FK: (BookTitle, PublisherName, YearOfPublication)\ references$
$BOOKPRICE3NF(BookTitle, PublisherName, YearOfPublication)$

Since the relational table AUTHORCATALOG2NF has no transitive dependency, it is qualified to be in 3NF. Hence,

$AUTHORCATALOG3NF(CatalogNum, AuthorFirstName, AuthorLastName)$
$PK: (CatalogNum, AuthorFirstName, AuthorLastName)$
$FK: (CatalogNum)\ references\ BOOKCATALOG3NF(CatalogNum)$

Since the 3 relational tables do not have non-trivial functional dependency, the 3 relational table are also in BCNF.

The final relational tables in BCNF are:

$BOOKPRICEBCNF(BookTitle, PublisherName, YearOfPublication, Price)$
$PK: (BookTitle, PublisherName, YearOfPublication)$

$BOOKCATALOGBCNF(CatalogNum, BookTitle, PublisherName, YearOfPublication)$
$PK: CatalogNum$
$FK: (BookTitle, PublisherName, YearOfPublication)\ references$
$BOOKPRICEBCNF(BookTitle, PublisherName, YearOfPublication)$

$AUTHORCATALOGBCNF(CatalogNum, AuthorFirstName, AuthorLastName)$
$PK: (CatalogNum, AuthorFirstName, AuthorLastName)$
$FK: (CatalogNum)\ references\ BOOKCATALOGBCNF(CatalogNum)$

**Deliverables**
A file solution1.pdf with the outcomes of the steps (i), (ii), and (iii) listed above. Note, that "educated guesses" of the solutions score no marks. You must provide the complete justifications of your answers.

Submission of a file with a different name and/or different extension and/or different type scores no marks!

**Task 3 (4.0 marks)**
**Functional dependencies and normalization**

A university records information about courses, instructors, and textbooks in a single table:

- Each **course** has a unique course code
- A course can be taught by multiple **instructors**
- A course can use multiple **textbooks**
- Instructors and textbooks are independent of each other (any instructor can use any textbook for the course)

**COURSE_INFO(**course_code**,** course_name**,** instructor**,** textbook_isbn**,** textbook_title**)**

**Sample Data:**

- ('CSCI235', 'Database Systems', 'Dr. Smith', '978-111111', 'Database Design Principles')
- ('CSCI235', 'Database Systems', 'Dr. Smith', '978-222222', 'Database Management Systems')
- ('CSCI235', 'Database Systems', 'Dr. Johnson', '978-111111', 'Database Design Principles')
- ('CSCI235', 'Database Systems', 'Dr. Johnson', '978-222222', 'Database Management Systems')

**Tasks for the Student:**

1. Identify the multi-valued dependencies and explain why this table violates 4NF
2. Normalize to 4NF relations

**Deliverables**

Submit a pdf file consisting of the normalization process and explanation of the processes of the above relational tables. Name your pdf file as **solution3.pdf**.

Sample solution includes but is not limited to the following:

1. Identify the multi-valued dependencies and explain why this table violates 4NF.

   **COURSE_INFO(**course_code**,** course_name**,** instructor**,** textbook_isbn**,** textbook_title**)**

   - Understanding the table and dependencies
     - $course\_code \rightarrow course\_name$ (Each course has a unique code and name).
     - A course ($course\_code$) can have multiple instructors.

    - A course ($\boldsymbol{course\_code}$) can have multiple textbooks ($\boldsymbol{textbook\_isbn, textbook\_title}$)
    - Instructors and textbooks are independent of each other. This means the choice of instructor for a course does not determine the textbook used, and vice-versa.
- From the above understanding, we can conclude that:
    - $\boldsymbol{course\_code \rightarrow course\_name}$
    - $\boldsymbol{course\_code \twoheadrightarrow instructor}$
    - $\boldsymbol{course\_code \twoheadrightarrow textbook\_isbn,\ textbook\_title}$

2. Normalize to 4NF relations.

   The solution to a 4NF violation caused by independent MVDs is to decompose the original table into separate relations, each representing one of the multi-valued dependencies. For the scenario described in this assignment, we need to decompose the COURSE_INFO relation into three relations:

- COURSE: Holds the core course information (functional dependency $\boldsymbol{course\_code \rightarrow course\_name}$.)

    - $\boldsymbol{COURSE(courseCode, courseName)}$
      **Primary key:** $\boldsymbol{courseCode}$

- COURSE_INSTRUCTOR: Holds the relationship between courses and instructors (first MVD).
    - $\boldsymbol{COURSE\_INSTRUCTOR(courseCode, instructor)}$
      **Primary key:** $\boldsymbol{(courseCode, instructor)}$
      **Foreign key:** $\boldsymbol{courseCode}$ **references COURSE(courseCode)**

- COURSE_TEXTBOOK: Holds the relationship between courses and textbooks (second MVD).
    - $\boldsymbol{COURSE\_TEXTBOOK(courseCode, textbookISBN, textbookTitle)}$
      **Primary key:** $\boldsymbol{(courseCode, textbookISBN)}$
      **Foreign key:** $\boldsymbol{courseCode}$ **references COURSE(courseCode)**

## Submissions

This assignment is due on Sunday, 26 April 2026 by 9:00 pm Singapore time.

Submit the files **solution1.pdf**, **solution2.pdf**, and **solutions3.pdf** through Moodle in the following way:

1) Zip all the files (Solution1.pdf, solution2.pdf, and solution3.pdf into one zipped folder.)
2) Access Moodle at **http://moodle.uowplatform.edu.au/**
3) To log in use a Login link located in the right upper corner the Web page or in the middle of the Web page
4) When successfully logged in, select a site CSCI235 (SP226) Database Systems
5) Scroll down to a section Submissions of Assignments
6) Click at Submit Your Assignment 1 here link.
7) Click at the button Add Submission
8) Move the zipped file created in Step 1 above into an area provided in Moodle. You can drag and drop files here to add them. You can also use a link *Add…*
9) Click at a button Save changes,
10) Click at check box to confirm authorship of a submission,
11) When you are satisfied, remember to click at a button Submit assignment.

**A policy regarding late submissions is included in the subject outline. Only one submission per student is accepted.**

Assignment 1 is an individual assignment, and it is expected that all its tasks will be solved individually without any cooperation with the other students. Plagiarism is treated seriously. Students involved will likely receive zero. If you have any doubts, questions, etc. please consult your lecturer or tutor during lab classes or over e-mail.

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*End of specification*